PHAS0050 : Climate & Energy
Module 3 : Climate Intervention

Prof. D. Waters, Prof. C. Ghag & Prof. C. Howard
Department of Physics & Astronomy, UCL

Contents

1 Introduction **2**

2 Carbon Dioxide Capture and Storage (CCS) **2**
2.1 The Fundamental Energy Cost of CO_2 Separation . . . 2
2.2 The Energy Generation Penalty of CCS . . . 4
2.3 Carbon Capture Techniques . . . 5
2.4 Carbon Storage Techniques . . . 7

3 Carbon Dioxide Removal (CDR) Techniqes **7**
3.1 CO_2 Capture From the Air . . . 8
3.2 Bio-Energy & Afforestation . . . 9
3.3 Enhanced Weathering . . . 9
3.4 Ocean Fertilisation . . . 9

4 Solar Radiation Management (SRM) Techniques **10**
4.1 Space Reflector . . . 10
4.2 Surface Albedo Enhancement . . . 10
4.3 Cloud Albedo Enhancement . . . 11
4.4 Stratospheric Aerosols . . . 13

5 Summary **13**

1 Introduction

This section of the course deals with Plan B : if greenhouse gas emissions cannot be abated sufficiently to avoid damaging or possibly catastrophic climate change, then what interventions can be considered to counteract or mitigate the effect of these emissions. Climate intervention could therefore be a long-term planned activity designed to reduce the impact of future emissions, or alternatively it could be envisaged as a rapid response in the event of some sort of "climate emergency". Individual climate intervention proposals span a wide range of cost, effectiveness and risk and may therefore lend themselves specifically to one or other of these deployment scenarios.

Section 2 deals with CO_2 capture and storage. While this is not strictly a climate intervention technique, many of the fundamental considerations are pertinent to the more general discussion of CO_2 removal techniques in section 3. Techniques for managing the atmospheric temperature directly by reducing the incident solar radiation are discussed in section 4. The various climate intervention techniques are summarised and compared in section 5.

2 Carbon Dioxide Capture and Storage (CCS)

There are a large number of proposals for capturing carbon dioxide and removing it from the climate system. There is a fundamental thermodynamic cost of separating CO_2 from a diffuse mixture which is first derived. Carbon capture is most straightforward at large point-sources such as power stations, and some of the important proposals are presented here. How to safely and economically store the large quantities of collected CO_2 outside of the climate system is also a major consideration.

Table 1 lists some of the important properties of CO_2 dioxide which are used at various point throughout this module. A phase diagram of CO_2 is shown in figure 1.

CO_2 Properties	
Molar mass	44
Atmospheric abundance	420 ppmv (parts per million by volume) 650 ppmw (parts per million by weight)
Temperature of sublimation at atmospheric pressure	$-78°\mathrm{C}$
Pressure of condensation at room temperature	50 bar

Table 1: Some properties of carbon dioxide.

2.1 The Fundamental Energy Cost of CO_2 Separation

There is a fundamental lower bound on the energy required to separate pure CO_2 from a gas mixture, which can be derived from basic thermodynamic considerations. The desired outcome, illustrated in figure 2, is the separation the CO_2 from other gases at constant pressure. After separation, the CO_2 and residual gas remain at the initial pressure and therefore occupy volumes in proportion to their relative abundance. In the case of CO_2 removal from the atmosphere, the CO_2 will occupy 0.039% of the total volume after separation (see table 1).

There is no change in temperature or internal energy of the separated components. However there is a decrease in entropy of the system - qualitatively, the separated components are in a state of reduced disorder with respect to the initial mixture. Now, reducing the entropy by an amount ΔS requires an amount of work equal to $-T \times \Delta S$, where T is the constant temperature maintained during separation. For an ideal gas at

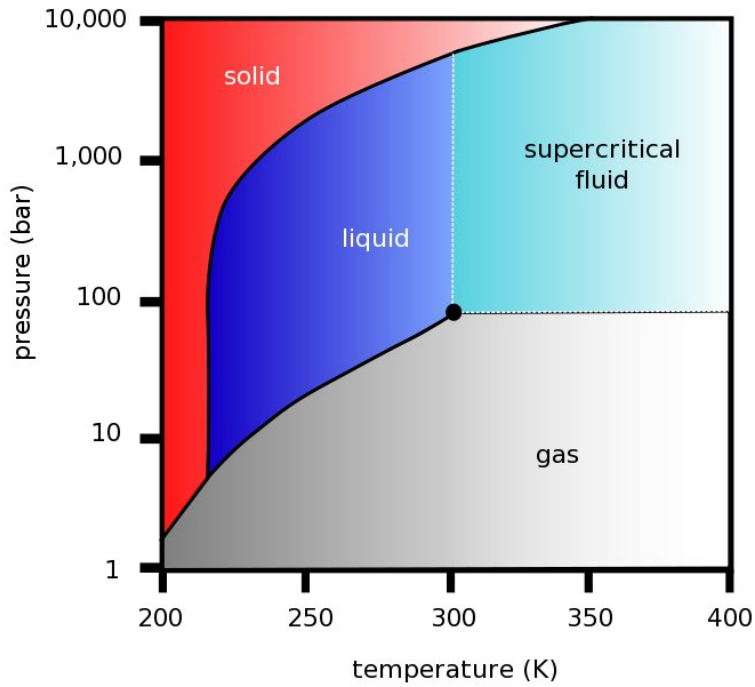

Figure 1: A phase diagram of CO_2, showing the state of this compound at a given temperature and pressure.

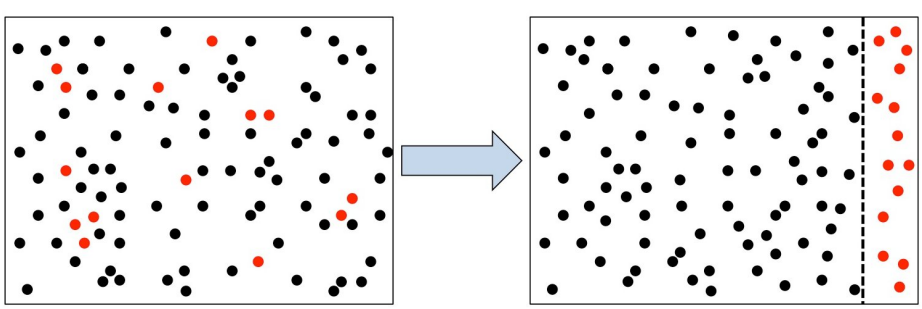

Figure 2: Separation of a mixture of ideal gases into components. The volume of each component after separation is proportional to the fractional abundance in the initial mixture. The total volume is unchanged.

constant temperature, the entropy of N atoms in a volume V is given by :

$$S = Nk \ln \left(\frac{V(kT)^{\alpha}}{CN} \right) \ , \tag{1}$$

where C is a constant (for a derivation, see for example equation 5.23c of [1]. Note that this formula is for a monatomic gas, but the rotational and vibrational degrees of freedom give rise to additional terms that do not depend on V.) Given that k, T, C and N are not varying, the entropy difference is simply due to the change in volume occupied by the CO_2 and is therefore given by :

$$\Delta S = S_f - S_i = Nk \ln \left(\frac{V_f}{V_i} \right) \ , \tag{2}$$

where V_i (V_f) are the initial (final) volumes occupied by the CO_2 respectively. Note that this entropy change is negative as expected. The work required per molecule of CO_2 is therefore :

$$W^{\text{separation}}_{1 \text{ molecule } CO_2} = -T \times \Delta S/N = kT \ln \left(\frac{V_i}{V_f} \right) \ . \tag{3}$$

Then the energy required to separate out 1 kg of CO_2 at an average atmospheric temperature of 14°C is :

$$W^{\text{separation}}_{1 \text{ kg } CO_2} = \frac{R \times 1000}{44} \times 287 \times \ln(100/0.039) = 425 \text{ kJ} = 0.12 \text{ kWh} \tag{4}$$

where the gas constant $R = 8.31$ J/mol/K. Clearly the required separation energy is smaller if the starting mixture contains a higher CO_2 concentration, as would be the case in the flue stream from a power station for example. Note that it is theoretically possible to separate CO_2 into a lower pressure state with no work required (for example, see the thought experiment in [1]). However one would then pay an equivalent energy penalty in compressing the CO_2 back to atmospheric pressure.

Indeed, the sequestration of carbon dioxide in geological formations or in the deep ocean requires further compression of the CO_2 to high pressure and this also has an associated energy cost. For example it is straightforward to show that isothermal compression from an initial pressure P_i to a final pressure P_f requires an energy :

$$W^{\text{compression}} = NkT \ln \left(\frac{P_f}{P_i} \right) \ . \tag{5}$$

Compressing CO_2 until it condenses at atmospheric temperature therefore requires :

$$W^{\text{compression}}_{1 \text{ kg } CO_2} = \frac{R \times 1000}{44} \times 287 \times \ln(50) = 212 \text{ kJ} = 0.06 \text{ kWh} \ . \tag{6}$$

In practice carbon dioxide gas is compressed to higher pressures, 100 atmospheres or more, prior to sequestration. Combining the results (4) and (6) above, the fundamental minimum energy required to capture CO_2 gas from the atmosphere and store it is approximately 0.2 kWh/kg. Techniques that are stated to require less energy must have an implicit input of energy from elsewhere - for example by employing exothermic chemical reactions as part of the capture process. In such a case the capture and storage is not operating in a closed cycle and a continuous supply of reactants is needed to sustain the sequestration (see for example the discussion in [3]).

2.2 The Energy Generation Penalty of CCS

The electrical energy yield from burning a certain mass of a fossil fuel source is E_{elec} and the energy required to capture and sequester the resulting CO_2 is E_{capture}. The difference, which is the useful energy supplied to the grid, is then :

$$E_{\text{supply}} = E_{\text{elec}} - E_{\text{capture}}$$

Fuel	Specific Carbon Content ($\mathrm{kg_C/kg_{fuel}}$)	Specific Energy Content ($\mathrm{kWh/kg_{fuel}}$)	Specific CO_2 Emission ($\mathrm{kg_{CO_2}/kg_{fuel}}$)	Specific CO_2 Emission ($\mathrm{kg_{CO_2}/kWh}$)
Coal	0.75	7.5	2.3	0.37
Natural Gas (Methane)	0.75	12	2.8	0.23

Table 2: Emission intensities of some fossil fuels. Note that the energies are thermal, not electrical (kWh, not $\mathrm{kWh_e}$).

The mass of CO_2 that is generated is denoted M_{CO_2}. Then writing $E_{\mathrm{elec}} = M_{CO_2} \times Y$, where Y is the electrical energy yield per kg of CO_2 and $E_{\mathrm{capture}} = M_{CO_2} \times C$, where C is the energy required to capture and sequester $1\ \mathrm{kg}$ of CO_2, we have :

$$E_{\mathrm{supply}} = M_{CO_2}\,(Y - C)$$

If no capture and storage takes place then $E_{\mathrm{supply}}^{\mathrm{no\ capture}} = M_{CO_2} \times Y$. Hence the energy generation penalty, which represents the additional mass of fuel that would be needed to provide the same useful energy when carbon capture is performed, is :

$$P_{\mathrm{CCS}} = \frac{E_{\mathrm{supply}}^{\mathrm{no\ capture}}}{E_{\mathrm{supply}}} = \frac{Y}{Y - C}$$

This makes sense : when the energy required to capture and store $1\ \mathrm{kg}$ of CO_2 becomes larger than the electrical energy liberated upon generating the same amount of gas, then CCS is impossible.

What is the penalty numerically ? Table 2 lists the typical emission intensities of some popular fossil fuels. It can be seen that the yield for burning coal is $1/0.37 = 2.7\ \mathrm{kWh/kg_{CO_2}}$. However this is the thermal yield, so assuming an efficiency of 40% for converting that energy into electricity, we have $Y = 2.7 \times 0.4 = 1.08\ \mathrm{kWh_e/kg_{CO_2}}$. Assuming $C = 0.2\ \mathrm{kWh/kg_{CO_2}}$, this gives a value $P_{\mathrm{CCS}} = 1.23$. In other words it would theoretically require the burning of approximately 20-25% more fuel in order to generate the same amount of useful energy and recover the entire corresponding amount of CO_2 from the atmosphere. It is important to point out that this is unrealistic for two reasons :

- Firstly, it would make no sense to generate more energy through burning fossil fuels in order to capture CO_2 at atmospheric concentrations, when it is typically available at much higher concentrations in the power plant itself. This carbon capture *at source* is the main focus of current emission reduction activity and is discussed in more detail in the next section. Of course scrubbing carbon dioxide directly from the atmosphere could itself be important if, for example, we wished to compensate for non-point like sources such as aviation, or if we wished to lower current atmospheric CO_2 concentrations to earlier or pre-industrial levels.

- Secondly, the above theoretical estimate for the minimum energy required to capture and sequester CO_2 is far from the current practical reality, potentially by a large factor.

There exists a wide range of estimates for the realistic energy (and therefore, to first order, the cost) penalty of capturing and sequestering CO_2 at source from fossil-fuel burning power plants, anything in the range of 10-50%.

2.3 Carbon Capture Techniques

Figure 3 indicates the general strategies for capturing CO_2 at source :

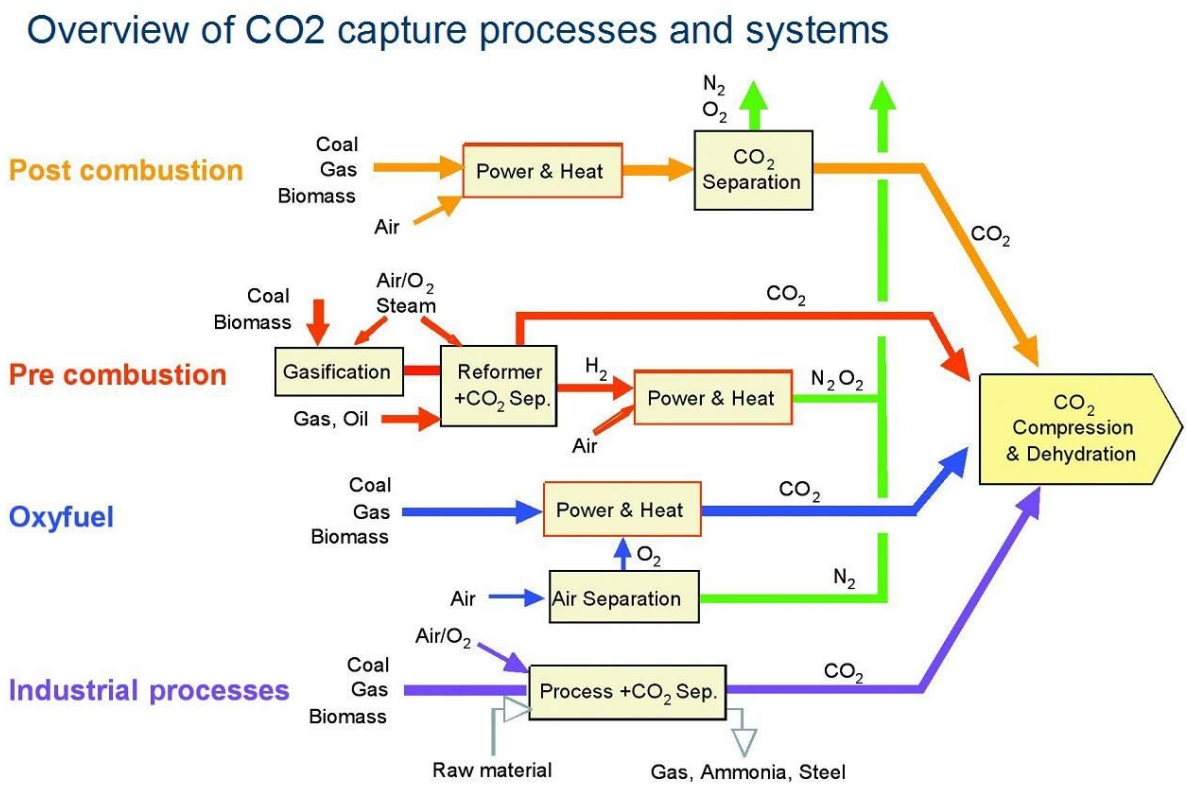

Figure 3: Possible CO_2 capture strategies from large point sources such as power stations and industrial plants. From [4].

- Post-combustion : the CO_2 is removed from the flue gasses after normal combustion, usually by absorption in a chemical solvent. The chemical solvent is then treated in some way (perhaps just by heating) to retrieve the captured CO_2 before being recycled back into the absorber column. This method is suitable (although very expensive) for retrofitting onto existing power stations. New-build power stations designed with post-combustion CO_2 capture in mind can provide more efficient capture - for example from the flue gasses at higher pressure.
- "Oxy-fuel" : oxygen is first separated from the air before combustion. Hence what remains after combustion in pure O_2 is a relatively pure mixture of CO_2 and H_2O. After drying the resulting gas can be sent directly to storage, implying an almost truly "zero-emission" process. On the other hand there are technical challenges, such as dealing with the very high temperatures encountered in combustion in a pure oxygen environment.
- "Pre-combustion" : a pre-combustion or "gasification" stage processes the primary fuel into separate streams of CO_2 for storage and H_2 which is used as a fuel in a power-generating turbine. An example of a pre-combustion technology is the Integrated Gasification Combined Cycle (IGCC) design. The capital costs for such power-plants are very high; one major advantage is that they could be a source of H_2 for transport or other applications.
- "Industrial Processes" : industrial processes such as the removal of CO_2 from natural gas, steel production and cement production are all important point-sources of CO_2 production which can be adapted to accommodate CO_2 capture through various means.

2.4 Carbon Storage Techniques

- "Geological Storage" : Storage of CO_2 at high pressures in suitable underground geological formations is currently the best understood possible scenario for sequestering carbon dioxide. Much of the technology exists, since very similar techniques are already used for the purposes of enhanced oil recovery in underground strata. There are some concerns about how stable such deposits might be over long time scales, but at least in certain geological formations it seems highly likely that the CO_2 would remain trapped for a very long period of time, possibly millions of years.
- "Ocean Storage" : At large depths liquid CO_2 is stable, denser than and relatively immiscible with the surrounding water. Figure 4 shows a beaker of liquid CO_2 at a depth of 3600 m. It may therefore be feasible to deposit CO_2 in giant undersea "lakes". This is relatively straightforward to realise technically but there are serious concerns about the long-term stability and effect on local ecosystems of such deposits.
- "Mineral Carbonation & Industrial Uses" : The "fixing" of CO_2 in mineral form is essentially identical in concept to the enhanced weathering techniques discussed in section 3.3. There are also various industrial uses of CO_2, some of which could become more economically viable in the event of plentiful supplies of captured carbon dioxide.

3 Carbon Dioxide Removal (CDR) Techniqes

As discussed in detail in section 2, the capture of carbon dioxide at large point-sources such as power stations is the most feasible method of making a significant reduction in greenhouse gas emissions from the burning of fossil fuels. However there are a multitude of smaller emitters for which carbon capture at source would not be feasible; moreover it may become necessary to actively reduce existing atmospheric CO_2 concentrations rather than just limit further emissions. The techniques discussed in this section are therefore aimed at removing CO_2 directly from the atmosphere.

Figure 4: A beaker of liquid CO_2 at a depth of 3600 m. From Monterey Aquarium Research Institute.

3.1 CO_2 Capture From the Air

Is it feasible to construct and operate a sufficient number of "scrubbers" to remove CO_2 directly from the air at the same rate that fossil fuel use is pumping CO_2 into the atmosphere ? Very simple arguments, for example outlined in [5], suggest that it is indeed feasible and may not be prohibitively expensive.

The density of air at ground level is approximately $1.2\ \mathrm{kg/m^3}$. Given the fraction of CO_2 in the air (see table 1), this corresponds to about $0.71\ \mathrm{g_{CO_2}/m^3}$ of CO_2. This density of carbon dioxide corresponds to an energy density :

$$\rho^E_{\mathrm{CO_2}}\ (\mathrm{J/m^3}) = \rho_{\mathrm{CO_2}}\ (\mathrm{kg/m^3}) \times I_{\mathrm{CO_2}}\ (\mathrm{J/kg})$$

where the emission intensity I is given in table 2. Taking the value for coal burning we find $\rho^E_{\mathrm{CO_2}} \sim 7000\ \mathrm{J/m^3}$. This can be compared to the kinetic energy density of air moving at, say, $10\ \mathrm{m/s}$ which is only $\frac{1}{2}mv^2 \sim 0.5 \times\ 1.2 \times\ 10^2 = 60\ \mathrm{J/m^3}$. What does this imply ? Burning fossil fuels and collecting the corresponding atmospheric CO_2 would require contact with a volume of air 2 orders of magnitude smaller than that required to generate the same amount of energy using wind turbines. Since it seems to be feasible to generate at least a few percent of our energy requirement from wind power, this suggests that the logistics of CO_2 capture from the air are not insurmountable.

More concretely : how many carbon capture devices might be required ? Anthropogenic CO_2 emissions are about $25\ \mathrm{Gt/yr}$ or approximately $4\ \mathrm{t/yr/person}$. A very modest average flow of $1\ \mathrm{m/s}$ through a CO_2 capturing device with 50% efficiency would result in a yield of approximately $10\ \mathrm{t/yr}$. Hence every person would need a carbon capturing device presenting a surface area to the wind of less than $1\ \mathrm{m^2}$. Again, at first glance this seems to be a potentially manageable figure.

Many physical and chemical absorption processes are known that are potential candidates for capturing CO_2 and then, in a second stage, releasing the captured CO_2 in a controlled way for sequestration. One candidate sequence of chemical reactions is the following :

$$\begin{aligned}
2\mathrm{NaOH} + \mathrm{CO_2(air)} &\rightarrow \mathrm{Na_2CO_3} + \mathrm{H_2O} \\
\mathrm{Na_2CO_3} + \mathrm{Ca(OH)_2} &\rightarrow 2\mathrm{NaOH} + \mathrm{CaCO_3} \\
\mathrm{CaCO_3} + \mathrm{HEAT} &\rightarrow \mathrm{CaO} + \mathrm{CO_2(sequester)} \\
\mathrm{CaO} + \mathrm{H_2O} &\rightarrow \mathrm{Ca(OH)_2}
\end{aligned}$$

Capture could take place in numerous relatively small-scale structure (see figure 5) or fewer larger structures, perhaps on the scale of power-station cooling towers. Despite the basic science being well understood

Figure 5: Devices to capture CO_2 directly from the atmosphere, sometimes called artificial trees.

there remain very significant technical challenges to be solved in order to make CO_2 capture from the air economically viable.

3.2 Bio-Energy & Afforestation

A number of carbon dioxide removal proposals involve the use of vegetation to fix atmospheric carbon dioxide. Trees or fast-growing crops could be grown and the carbon-rich organic material either sequestered (i.e. buried) directly, or more likely burned as fuel in a power station equipped with CCS.

Such schemes are relatively easy to implement and very low risk; the key difficulty is the large implications for land-use. The most efficient crops can only fix about 1500 tons of CO_2 per km^2 per year. This would imply devoting something like an additional 10% of the Earth's land area to carbon dioxide capture if there is no reduction in current CO_2 emissions.

3.3 Enhanced Weathering

Weathering is a natural process which removes CO_2 from the atmosphere through reactions with carbonate and silicate rocks. An example chemical reaction is the following :

$$\mathrm{CaSiO_3} + \mathrm{CO_2} \rightarrow \mathrm{CaCO_3} + \mathrm{SiO_2}$$

Eventually such processes will naturally remove excess CO_2 from the atmosphere. The problem is that the reaction rate is extremely small and the challenge is therefore to accelerate these processes. For example certain kinds of minerals could be ground-up and incorporated into agricultural practices such that they cover large surface areas. In any event the masses of minerals involved and corresponding logistics are formidable and probably expensive.

3.4 Ocean Fertilisation

Algal photosynthesis directly fixes CO_2 in surface waters and subsequently by absorption from the atmosphere. A certain fraction of the carbon sinks to greater depths and is therefore removed from the climate system for a considerable period of time.

The key suggestion behind ocean fertilisation proposals is that algal photosynthesis and certain other carbon-fixing biological processes in surface waters are often limited by the availability of specific nutrients. It may only require the delivery of relatively small masses of nutrients to stimulate biological activity capable of capturing much larger masses of CO_2. There are however serious concerns about what the unpredictable consequences of such major interventions in ocean eco-systems might be.

4 Solar Radiation Management (SRM) Techniques

The basic idea behind solar radiation management is to directly compensate for the additional radiative forcing due to greenhouse gases in the atmosphere by reducing the amount of solar energy absorbed into the climate system.

If S_0 is the average solar irradiance (in units of $\mathrm{W/m^2}$) at the top of the atmosphere and R_E is the radius of the Earth, then the average solar irradiance F_S received at the Earth's surface is given by :

$$\begin{aligned} F_S &= \frac{\text{solar power incident on Earth}}{\text{surface area of Earth}} \times \text{fraction absorbed} \\ &= \frac{S_0 \times \pi R_E^2}{4\pi R_E^2} \times (1-a) = \frac{S_0}{4}(1-a) \ , \end{aligned} \tag{7}$$

where $R_E = 6371\ \mathrm{km}$ is the radius of the Earth and a is the average albedo of the Earth, or the fraction of incident energy reflected harmlessly back into space. Numerically, $S_0 = 1370\ \mathrm{W/m^2}$ and $a \sim 0.3$, hence :

$$F_S = \frac{1370}{4}(1-0.3) = 240\ \mathrm{W/m^2}$$

The radiative forcing due to a doubling of atmospheric CO_2 corresponds to approximately $4\ \mathrm{W/m^2}$. Hence to balance this warming effect, the incident solar radiation must be decreased by approximately $4/240 \sim 1.7\%$. The various possible strategies for achieving this are indicated schematically in figure 6 and discussed in further detail below.

4.1 Space Reflector

Positioning a reflector between the Sun and Earth would act to reduce S_0 in equation 7 above. The logistics are formidable, if not overwhelming. To achieve the 1.7% reduction in incident solar flux required, then over 2 million $\mathrm{km^2}$ of mirrors or other kind of shade (e.g. orbiting dust particles) would be required. Rather than structures orbiting the Earth, another possibility is to place material at much larger distances such as at Lagrange point L1 between the Sun and Earth. At such a large distance from Earth, light would only have to be refracted through a small angle to avoid impinging upon the planet.

4.2 Surface Albedo Enhancement

Changing the average reflectivity of the Earth's surface would serve to reduce the fraction of the incident solar energy that enters the climate system. A number of proposals have been made which all have serious drawbacks : for example increasing the albedo of urban areas (painting roofs white) would be relatively straightforward to achieve on a long time-scale (e.g. by changing building regulations) but is very unlikely to add up to a large enough effect. Increasing the albedo of deserts with the placement of very large areas of reflective material could give a more sizeable effect but could potentially give rise to significant and poorly understood changes to the local climate.

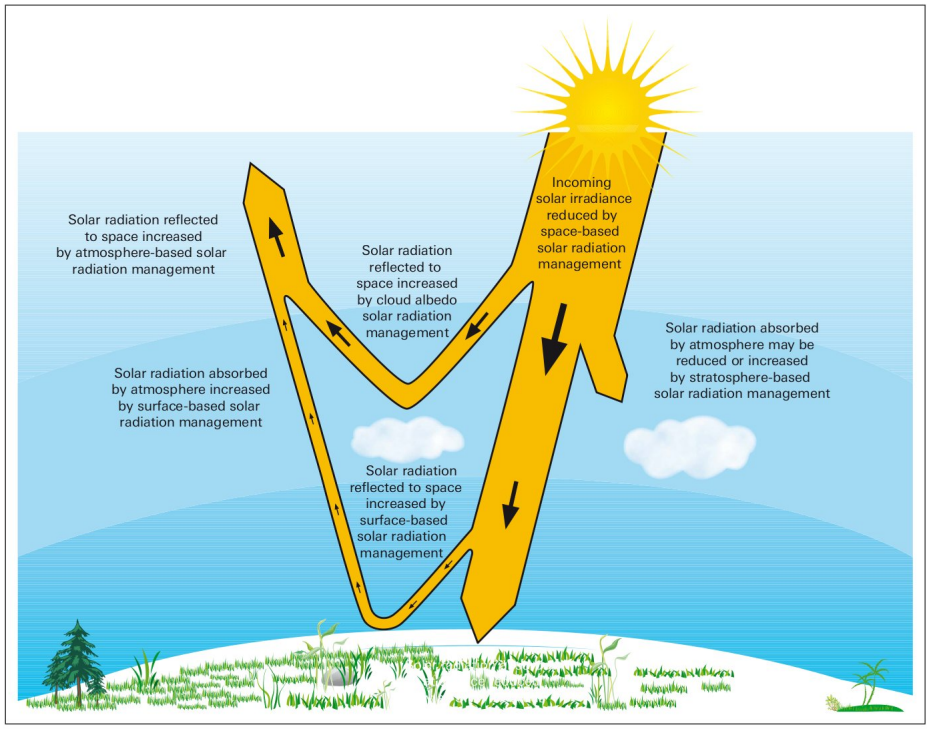

Figure 6: A schematic showing possible methods for achieving a reduction in the solar energy entering the climate system [from [2]].

4.3 Cloud Albedo Enhancement

Clouds form when droplets condense on tiny particles called cloud condensation nuclei (CCN). Suppose the initial number density of CCN is N_0. If this is increased to N, then the same mass of condensate will form a larger number of smaller droplets, with the ratio of radii being :

$$\frac{r}{r_0} = \left(\frac{N}{N_0}\right)^{-1/3}$$

The total surface area of droplets will therefore be increased by a factor :

$$\frac{A}{A_0} = \frac{N \times 4\pi r^2}{N_0 \times 4\pi r_0^2} = \frac{N}{N_0}\frac{r^2}{r_0^2} = \left(\frac{N}{N_0}\right)^{1/3}$$

The larger surface area of droplets will increase the cloud albedo to the incident sunlight; in other words the clouds will be "whitened" (although not necessarily by a factor directly proportional to the increase in droplet surface area). A further advantage is that smaller droplets typically have a longer residence time in the atmosphere before precipitating out.

By what amount do we need to increase the number of CCN to have the desired effect on the climate ? We take as an example proposals to increase the albedo of low-level clouds above the ocean [6]. From equation 7 above, we have :

$$\Delta F_S = -\frac{S_0}{4}\Delta a \approx -340 \times \Delta a \ \ \mathrm{W/m^2}$$

Hence to reduce the irradiance by $4\ \mathrm{W/m^2}$ requires an increase in the average planetary albedo of about 0.011, or about 3.9% relative to the $a = 0.3$ assumed above. Since oceans cover about 70% of the world's

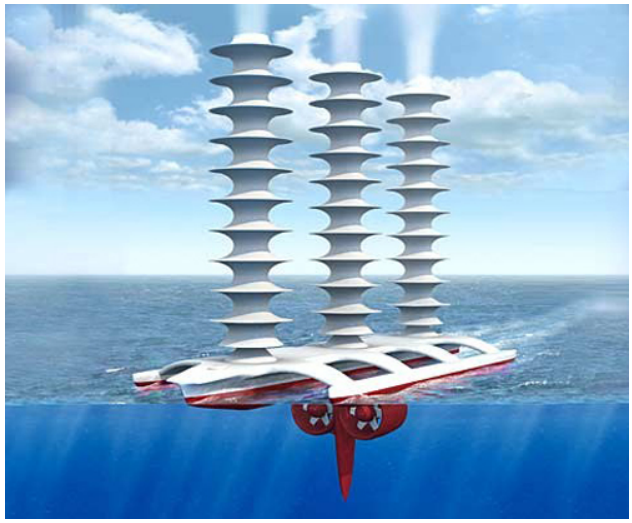

Figure 7: A sketch of an automatic vessel that could be used to deliver fine water droplets for the purposes of whitening low-lying marine cloud.

surface area and approximately 25% of that is covered by low-lying cloud at any one time, then those regions of the surface of the Earth need to increase their albedo by $\approx 0.011/(0.7 \times 0.25) = 0.063$. The albedo of these clouds is already about 50% and therefore an increase by 0.063 corresponds to about a 13% relative increase. It can be shown (for details see [6]) that for these clouds :

$$\Delta a = 0.075 \ln (N/N_0)$$

Hence $\Delta a = 0.063$ gives $N \approx 2.3 \times N_0$. In simple terms, one needs to roughly *double* the number of CCN to achieve the required albedo enhancement.

Droplets of sea-water with dissolved NaCl can provide the required CCN. An estimate of the rate at which sea-water droplets would need to be dispersed is the following :

$$\frac{dn}{dt} = \frac{A_E \times 0.7 \times 0.25 \times H}{\tau \times f} \times (N - N_0) \ ,$$

where A_E is the surface area of the Earth and the two numerical factors in the numerator take into account the fraction of the Earth's surface covered by low-level marine cloud as described above. H is the column height over which droplets would need to be dispersed and, importantly, τ is the residence time of the droplets in the atmosphere. The factor f in the denominator takes into account the fraction of droplets produced that successfully enter the cloud formation region. Reasonable values for these parameters are : $H = 1000$ m, $\tau = 3$ days and $f = 0.5$. Then given that $N_0 \sim 100 \ \mathrm{cm}^{-3}$, doubling the number of CCN requires an injection rate :

$$\frac{dn}{dt} \sim 7 \times 10^{19} \ \ \mathrm{s}^{-1}$$

If fine droplets with a radius $r_d = 0.5 \ \mu\mathrm{m}$ could be generated and dispersed, this corresponds to a volume ejection rate of :

$$\frac{dV}{dt} = \frac{dn}{dt} \times \frac{4}{3}\pi r_d^3 = 37 \ \ \mathrm{m}^3/\mathrm{s}$$

This appears not to be completely unfeasible. It has been suggested that such a droplet load could be delivered by a fleet of automatic ships, illustrated schematically in figure 7.

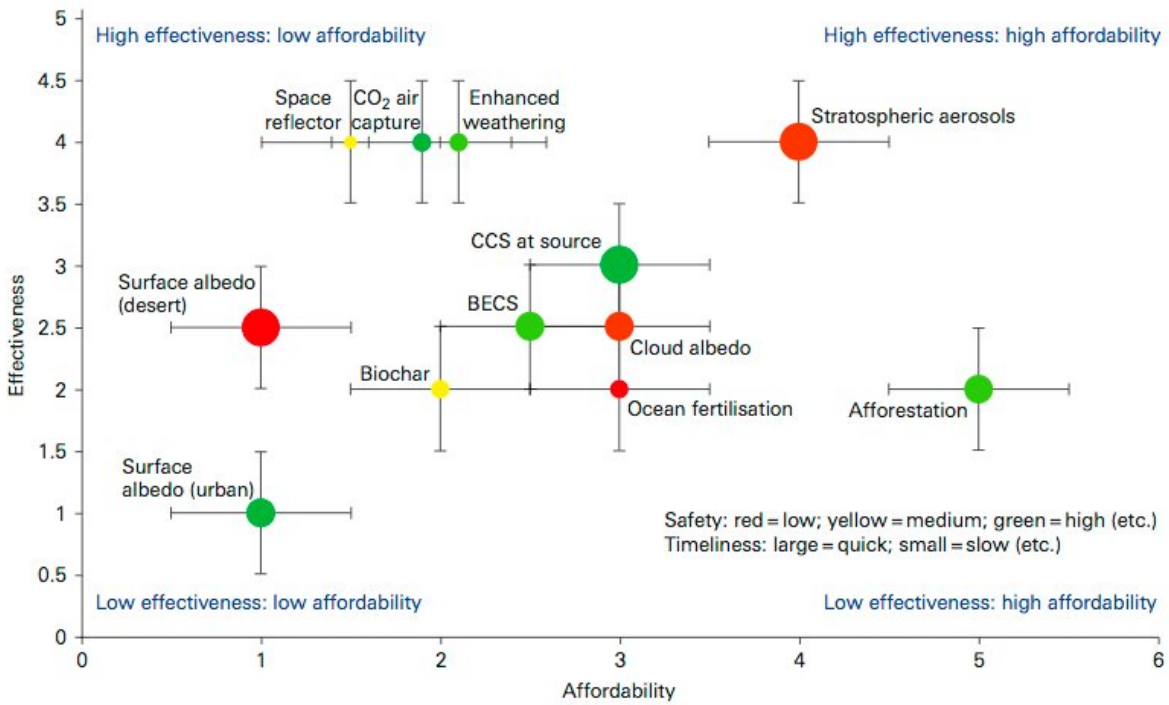

Figure 8: A comparison of the relative affordability and effectiveness of various proposed geo-engineering techniques [from [2]]. Note also the safety colour coding (red=high-risk; green=low risk) and the timeliness size variation (large=quick to implement; small=slow to implement).

4.4 Stratospheric Aerosols

A number of proposals fall under this category. Promoting high-level cloud formation, which already happens to some extent when aircraft leave "contrails" in the upper troposphere or lower stratosphere, is a possibility. A quantitative analysis would be similar in many respects to that developed in the previous section for low-lying marine cloud.

More directly, the injection of particles into the upper atmosphere can provide a sun-shading effect. Indeed this happens naturally from time to time through volcanic activity and is reasonably well understood. The effect is simple to calculate using equation 7 for a given particle size and abundance. Small sulphate particles, perhaps $0.1\ \mu\mathrm{m}$ radius, might be optimal. It is then estimated that between 1 and 5 $\mathrm{MtS/yr}$ (mega-tons of sulphur per year) would need to be injected into the stratosphere. Surprisingly transporting such quantities does not seem to be unfeasible, since it is on the same scale that commercial air transport already transfers mass at high altitudes.

5 Summary

The climate intervention methods discussed here, and more, are summarised in figure 8, taken from [2]. Not all of the surrounding issues of risk and timeliness have been explored above, but nevertheless a few clear trends are apparent :

- Carbon capture and storage, and other carbon dioxide removal (CDR) techniques, can be safe and rather effective. For example CO_2 air capture and enhanced weathering are simply the removal of excess carbon dioxide that we have placed in the atmosphere, returning to an earlier atmospheric con-

figuration. However not all carbon capture proposals are without risk - most notably ocean fertilisation which would produce major perturbations to local ecosystems.

- No solar radiation management (SRM) technique that is effective is without risk. The fundamental reason for this is that such techniques do not return the climate system to a previous configuration, rather they put the climate system in a new configuration in just such a way that two effects (a smaller solar irradiance and a larger greenhouse effect) cancel each other out to produce no net change in average temperature. But the new climate configuration may be different in many other respects - in particular local temperature and precipitation patterns might be quite different from the original unperturbed climate system. Such consequences may be very hard to predict reliably. The great advantage of some of the SRM techniques such as the use of stratospheric aerosols is that they are potentially very effective at rapidly reducing average temperatures in a reasonably affordable way.

As sections 2.1 and 2.2 make clear, fundamental physical considerations do not imply a large energy penalty associated with carbon dioxide capture and sequestration, even at atmospheric concentrations. However the realisation of economically viable technologies is still some way off; moreover the threshold of economic viability is partly determined by what are judged to be the potential costs of *not* intervening in the climate.

References

[1] "Statistical Physics", F. Mandl, 2nd edition, page 128.

[2] "Geoengineering the climate : Science, governance and uncertainty", Royal Society Policy Document 10/19 (2009).

[3] "Sustainable Energy - without the hot air", David MacKay (2009).

[4] "IPCC Special Report on Carbon Dioxide Capture and Storage", edited by B. Metz *et al.* (2005).

[5] K. Lackner, P. Grimes and H.-J. Ziock, "Capturing Carbon Dioxide from Air", First National Conference on Carbon Sequestration, National Energy Technology Laboratory (2001).

[6] J. Latham *at al.*, "Global temperature stabilization via controlled albedo enhancement of low-level maritime clouds", Phil. Trans. R. Soc. A **366**, 3969 (2008).